

MLDL2 Homework3: Transfer Learning

Deadline: June 17, 2026, by 11:59 PM

Goal:

Perform Unsupervised Domain Adaptation (UDA) using labeled source data and unlabeled target data. Starter code and data can be downloaded from [here](#).

Task Description:

UDA is a representative form of transfer learning. In real-world scenarios, the distribution of the source data may differ from that of the target data on which the model is expected to perform. For example, the source data may be well-curated, with a uniform number of samples per class and high-resolution images, whereas the target data may have class imbalance or lower image quality. **UDA aims to train a model to achieve robust performance on the target domain under such distribution shifts**, using labeled source data and unlabeled target data.

In this assignment, we provide two datasets: **CUB-200** and **CUB-200-Paintings**. You will perform unsupervised domain adaptation under the following two settings:

[Setting1] $C \rightarrow P$: Use CUB-200 as Source, and CUB-200-Paintings as Target.

[Setting2] $P \rightarrow C$: Use CUB-200-Paintings as Source, and CUB-200 as Target.

UDA assumes **access to labeled source and unlabeled target data**.

- **During training**, the model uses images with labels from source data and images without labels from target data.
- **During evaluation**, the trained model is tested on target images, and its predictions are compared with ground truth target labels.

You may use various training strategies, including supervised learning on the source domain, semi-supervised learning, unsupervised learning, data augmentation, hyperparameter tuning, architectural modifications, replay buffers, and other relevant techniques.

Datasets:

1. **CUB-200-2011 Dataset (Caltech UCSD Birds 200):**
 - CUB-200-2011 is a fine-grained bird classification dataset containing images from 200 bird species.
2. **CUB-200-Paintings Dataset:**

- CUB-200-Paintings is a dataset of bird paintings collected from the web and manually filtered. It shares the same class list as CUB-200-2011. The dataset includes various artistic styles, such as watercolors, oil paintings and pencil drawings.
- Unlike CUB-200, CUB-200-Paintings is smaller in size and has a more imbalanced number of samples across classes.

Evaluation:

Evaluate your model on the target data. **Do not modify the [Evaluate and Submit] section.** Running this section will save predictions_CtoP.npy and predictions_PtoC.npy, both of which must be included in your submission zip file.

Submission Guidelines:

Submit a .zip file to **ETL > DL_HW3** Assignment Submission. It must include

- `StudentID_Name.pdf`: the report in PDF format
- `StudentID_Name.ipynb` or `StudentID_Name.py`: your source code
- `predictions_CtoP.npy` and `predictions_PtoC.npy`: your prediction files

The report must be within 2 pages, excluding references, with a font size of at least 11 pt and line spacing of at least 1.15. It should clearly justify the chosen training methods and strategies based on the dataset characteristics and assignment constraints.

Grading & Important Notes:

The final score consists of **model performance (80 points)** and the **report (20 points)**.

[Model Performance]

- Pretrained weights and target labels are not allowed during training.
- Using target data itself is optional and will not be penalized; however, leveraging unlabeled target data is likely to improve performance.
- Model performance will be graded based on ranking:
Top 3: full credit / Up to Q3: 75 / Up to Q2: 60 / Up to Q1: 40 pts

[Reports]

- Reports exceeding the page limit will receive a **5-point penalty**. **Late submissions will not be accepted.**